ABIGAIL LARA BURNS

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Passionate and thorough geospatial professional dedicated to informing and empowering decision-making.

EDUCATION

Master of Science in Social Data Analytics and Research

The University of Texas at Dallas – School of Economic, Political and Policy Sciences

Bachelor of Science in Geospatial Information Sciences

The University of Texas at Dallas – School of Economic, Political and Policy Sciences

TECHNICAL SKILLS

Programming Languages:	Python, R, SQL, Arcade, VBA, HTML
Applications and Software:	Esri ArcGIS Pro, ArcGIS Enterprise, ArcGIS Online, ArcGIS Server, ERDAS Imagine, QGIS, Microsoft Excel, Microsoft SQL Server, SQLite
Functions:	Cartography, Spatial and Geostatistical Analysis, Scripting, App Development, Spatial ETL, Data Design

PROJECTS

BLE Post-Processing – Python, ArcPy, floodplain mapping and validation

Weather and News Scraping – Selenium, pandas, RSS, dictionaries, text mining, sentiment analysis, shiny apps

DART Red Line Schedule – SQLite, SQL database, table population and schema creation, shiny apps

WORK EXPERIENCE

GIS Specialist II – AECOM – Dallas, TX *Jan 2024 – Present*

- Use 2D hydraulic modeling to perform floodplain analysis as part of the Base Level Engineering approach.
- Provide flood risk and other spatial datasets to FEMA for unstudied or previously 1D-modeled communities.
- Document spatial analysis processes and best practices to meet Compass quality metrics.
- Develop Python scripting tools for raster analysis, data validation, and dataset creation.
- Develop Visual Basic macros for reporting building site inspections and energy usage.
- ➢ Automated workflows and improved functionality of existing spatial data processes.
- ➢ Collaborated across offices and business lines to meet client needs.

GIS Analyst – Town of Prosper Municipal Government – Prosper, TX *Aug 2022 – Nov 2023*

- Reviewed, digitized, and maintained information for ongoing land developments.
- Created utility pipeline data from georeferenced As Built and CAD drawings.
- Designed domains and schema for utility infrastructure, planning, and public safety datasets.
- Collected field data and configured services and apps for internal and public use.
- Administered SQL Server instances, users, and database replication.
- ➢ Improved accuracy and management of public utility assets.
- ➢ Effectively deployed and supported GIS capabilities for town departments and residents.

Contract Employee – NewGen Strategies & Solutions – Richardson, TX *Feb 2022 – May 2022*

- Performed reliably accurate supervised object-based image classification and analysis.
- Tabulated impervious areas to calculate Equivalent Residential Units.
- ➢ Created actionable data for stormwater runoff management and flood mitigation.

GIS Coordinator PT – Collin County Government – McKinney, TX *Mar 2021 – Aug 2022*

- Reverse-geocoded to assign 9-1-1 rural addresses and replicated and synchronized this data with NCTCOG.
- Automated spatial ETL of city transportation data using Python scripting and ModelBuilder.
- ➢ Efficiently and accurately addressed burgeoning property developments.